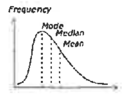
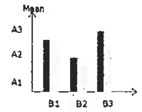
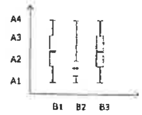
SPSS 2 Descriptive Statistics

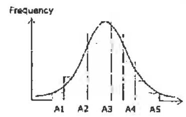
In this session

- What is descriptive statistics?
- Central tendency and Variability
- Generate statistics
  - Summary Table, Pie Chart, Cluster Bar Chart, Cluster Line Chart
  - OLAP Cube, Percentiles, Box Plot
- Normal Distribution







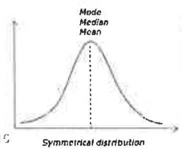


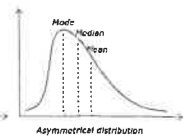
What is descriptive statistics?

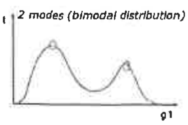
- Quantitatively summarizing main features of data without drawing conclusion
- Consists of : head, tail flat
  - Summary statistics: eg. central tendency, variability, skewness, kurtosis, correlation coefficient
  - Visual: eg. frequency plot, box plot
- Univariate (one-variable) or bivariate (multi-variable) analysis

Central Tendency

- Central tendency: find the **representative value** in the distribution
  1. Mode
     - The most frequent value
     - More representative when there are more than one peaks
  2. Median
     - The middle value that divide the distribution in 2 halves
     - More representative in an unbalanced distribution
  3. Mean
     - Average of all values
     - For unbalanced distribution , mean is at the tail representing minority
- For an unbalanced distribution, mean is always at the tail, mode is at the head, median in the middle



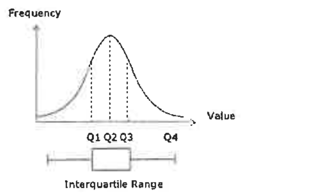




2 'mode's is useful than 1 mode

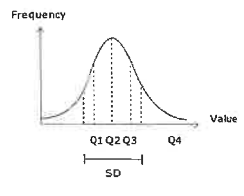
Variability

- **Range**
  - Difference between highest and lowest values
  - Unstable. Sensitive to extreme values and skewed distribution
- **Interquartile Range**
  - Difference between 1^{st} and 3^{rd} quartile
  - Range of the middle 50% distribution
  - Less sensitive to extreme values
- **Quartiles**
  - Divide the distribution in 4 equal-sized subsets
  - A value at the 3^{rd} quartiles means 3/4 of values are below it



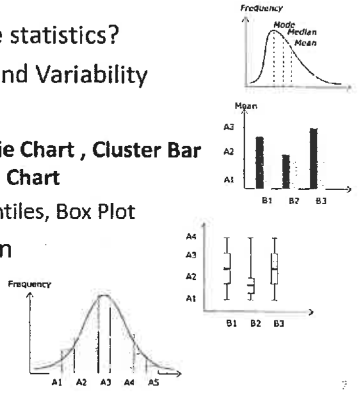
Variability (cont')

- **Standard Deviation** (SD)
  - Average difference between all values to the mean
  - Has a close relationship with the normal curve
- **Variance** (δ)
  - Variance = SD^2 $S.D. = \sqrt{\delta}$
  - Useful in doing certain calculation



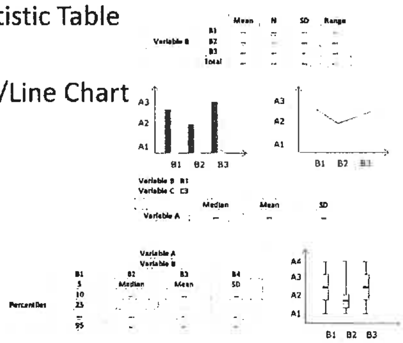
In this session

- What is descriptive statistics?
- Central tendency and Variability
- Generate statistics
  - **Summary Table, Pie Chart , Cluster Bar Chart, Cluster Line Chart**
  - OLAP Cube, Percentiles, Box Plot
- Normal Distribution



Central Tendency And Variability Statistics

1. Summary Statistic Table
2. Pie Chart
3. Clustered Bar/Line Chart
4. OLAP cube
5. Percentiles
6. Box Plot



Summary Statistic Table

- Generate central tendency and variability statistic
  - Divide the data into subsets based on a variable
  - Generate statistic of another variable for all subsets
- To generate summary statistic for *variable A* by *B:*
  1. Double click file "gssdata.sav" to open in SPSS
  2. **Analyze > Compare Means > Means**
  3. Make settings (see the red color text below)

Always take scale data to analyse

Summary statistic for A by B

Variable A (Dependent List)

Variable B (Independent List)	Mean	N	SD	Range
B1	...	...	...	...
B2	...	...	...	...
B3	...	...	...	...
Total	...	...	...	...

Range —— Options > Cell Statistics

Taking non-scale data is meaningless.

Task 1a – Summary Statistic Table

1. Use the previous slide to generate the following summary statistic table
2. What is the range of age of all people?
3. What is the SD of age of people that never read news paper?
4. What is the average age of people that read news paper every day?
5. What are the average ages of the most frequent and the least frequent readership group? What pattern do you find in the 2 groups?

Report

Age of respondent

How often do you read a newspaper?	Mean	Std. Deviation	[illegible]	Variance	Minimum	Maximum	Range	Std. Error of Mean
Everyday	52.60	[illegible]	367	316.230	19	89	[illegible]	.928
Few times a week	40.75	[illegible]	217	240.486	19	[illegible]	65	1.053
Once a week	41.72	16.557	108	274.128	20	87	67	1.593
Less than once a week	40.35	16.032	120	257.036	[illegible]	89	70	1.464
Never	[illegible]	18.169	88	330.126	[illegible]	89	69	1.937
Total	[illegible]	17.776	900	315.899	[illegible]	89	70	.593

Task 1a answer – Summary Statistic Table

- Summary statistic for *age* by *news reading habit:*
  1. Double click file "gssdata.sav" to open in SPSS
  2. **Analyze > Compare Means > Means**
  3. Set **Dependent List** to age, **Independent List** to news
  4. **> Options** > Set **Cell Statistics** to Mean, Median, Standard Deviation, Variance, Minimum, Maximum, Range, Std. Error Of Mean, Number Of Cases
  5. **> Continue > Ok**

Report

Age of respondent

How often do you read a newspaper?	Mean	Std. Deviation	N	Variance	Minimum	Maximum	Range	Std. Error of Mean
Everyday	52.60	17.783	367	316.230	19	89	70	.928
Few times a week	40.75	15.508	217	240.486	19	84	65	1.053
Once a week	41.72	16.557	108	274.128	20	87	67	1.593
Less than once a week	40.35	16.032	120	257.036	19	89	70	1.464
Never	45.99	18.169	88	330.126	20	89	69	1.937
Total	46.16	17.776	900	315.899	19	89	70	.593

Task 1a answer – Summary Statistic Table (cont')

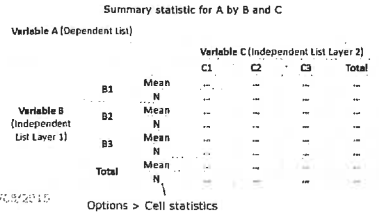
1. Generate the following summary statistic table
2. What is the range of age of all people?
   - 70
3. What is the SD of age of people that never read news paper?
   - 18.17
4. What is the average age of people that read news paper every day?
   - 52.6
5. What are the average ages of the most frequent and the least frequent readership group? What pattern do you find in the 2 groups?
   - Most frequent: 52.6, least frequent: 45.99, pattern: eldest groups

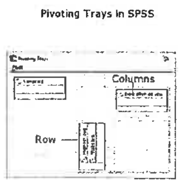
Age of respondent

How often do you read a newspaper?	Mean	Std. Deviation	N	Variance	Minimum	Maximum	Range	Std. Error of Mean
Everyday	52.60	17.783	367	316.230	19	89	70	.928
Few times a week	40.75	15.508	217	240.486	19	84	65	1.053
Once a week	41.72	16.557	108	274.128	20	87	67	1.593
Less than once a week	40.35	16.032	120	257.036	19	89	70	1.464
Never	45.99	18.169	88	330.126	20	89	69	1.937
Total	46.16	17.776	900	315.899	19	89	70	.593

Two-variable Summary Statistic Table

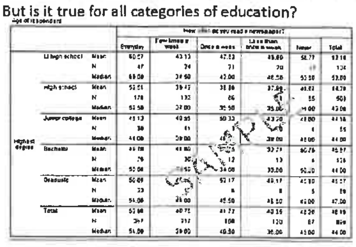
- Explore the data in more details
- Generate subset statistics for each combination of the two variables
- To generate summary statistic for *variable A* by *B* and *C*:
  1. **Analyze > Compare Means > Means**
  2. Make settings (see red color text below)
  3. **Double click** the table to edit its columns and rows in the Pivoting Trays





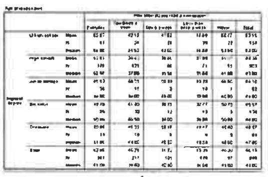
Task 1b – Two-variables Summary Statistic Table

1. Use the previous slide to generate the following summary statistic
2. What is the mean age of respondents that read newspaper once a week with the highest degree as Bachelor?
3. What is the median age of respondents with the highest degree as High School?
4. What is the total number of respondents that never read newspaper?
5. We already know from the previous task that people who read newspaper everyday are the oldest. But is it true for all categories of education?



Task 1b answer – Two-variables Summary Statistic Table

- Summary statistic for *age* by *newspaper reading habit* and *education*:
  1. **Analyze > Compare Means > Means**
  2. Set **Dependent List** to age, set **Independent List** to degree > **Layer** > **Next** > set **Independent List** to news
  3. **> Options** > Set **Cell Statistics** to Mean, Median, Number Of Cases > **Continue** > **OK**
  4. **Double click** the generated table > **right click** > **Pivoting Trays** > set **Row Dimension** to highest degree & statistics; set **Column Dimension** to how often do you read a newspaper?
  5. **Right Click** > **Table Property** > **General** > Set **Row Dimension Labels** to nested
  6. **Click** anywhere outside the table to exit the editor



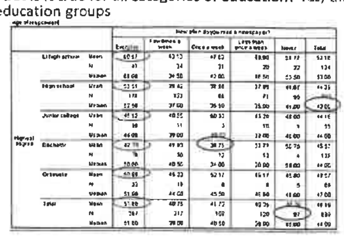
independent : Scale

Dependent : Non-scale

Task 1b answer – Two-variables Summary Statistic Table (cont')

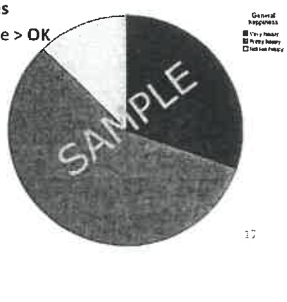
Based on the statistic:

1. Generate the following summary statistic
2. What is the mean age of people that read newspaper once a week with the highest degree as Bachelor? 39.75
3. What is the median age of people with the highest degree as High School? 43
4. What is the total number of people that never read newspaper? 87
5. We already know from the previous task that people who read newspaper everyday are the oldest. But is it true for all categories of education? Yes, the ages are above average in all education groups



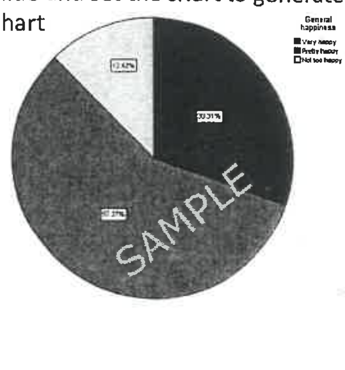
Pie Chart

- Pie chart is useful to show the relative frequencies of categorical data
- To generate pie chart based on frequencies of data:
  1. **Graphs > Legacy Dialogs > Pie**
  2. Select **Summaries of groups of cases > Define**
  3. Select **Slice Represent** > **N of cases**
  4. Set **Define Slices by:** to the variable > **OK**



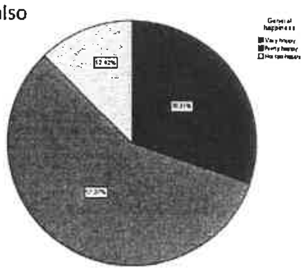
Task 2 – Pie Chart

1. Double click file "gssdata.sav" to open
2. Use the previous slide and set the chart to generate the following pie chart



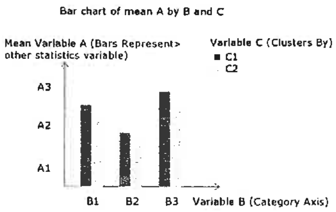
Task 2 answer – Pie Chart

1. **Graphs > Legacy Dialogs > Pie Chart**
2. Set **Slice By** to <u>happy</u> (which is also the *categorical variable*)
3. **Double click** chart to edit > **Elements > Show Data Labels > Data Value Labels >**
4. Set **Displayed List** to <u>Percent</u> > **Apply** and **Close**
5. **Close** the Chart Editor



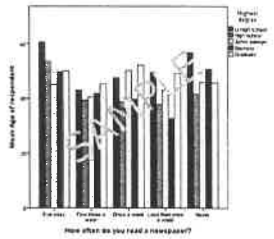
Clustered Bar Chart of Means

- Visualize statistics of all subsets of data
- To generate bar chart of *Variable A* by *B* and *C*
  1. **Graphs > Legacy Dialogs > Bar**
  2. Select **Clustered** and **Summaries For Group Of Cases > Define**
  3. Make setting (see red color text below)



Task 3a - Clustered Bar Chart of Means

1. Use the previous slide to generate the following bar chart
2. Which *news* and *degree* group has the highest mean *age* of respondents?
3. Which *degree* group that read newspaper once a week has the highest mean *age* of respondents?



Task 3a answer - Clustered Bar Chart of Means

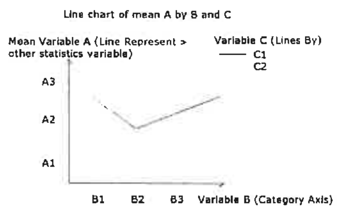
- Visualize statistics of all subsets of data
- Bar chart of average *age* by *news reading habit* and *education*
  1. **Graphs > Legacy Dialogs > Bar**
  2. Select **Clustered** and **Summaries For Group Of Cases > Define**
  3. set **Category Axis** to <u>news,</u>
  4. set **Define Clusters By** to <u>degree,</u>
  5. Set **Bars Represent** to <u>other statistics,</u>
  6. set **Variable** to <u>age</u> > **Ok**
  7. **Double Click** chart to edit > **Options > Y Axis Reference Line >** set **Position** to: <u>46.18</u>
  8. **> Apply > OK**

Task 3a answer - Clustered Bar Chart of Means (cont')

1. Use the previous slide to generate the following bar chart
2. Which *news* and *degree* group has the highest mean *age* of respondents?
   - Everyday and Lt high school
3. Which *degree* group that read newspaper once a week has the highest mean *age* of respondents?
   - Graduate

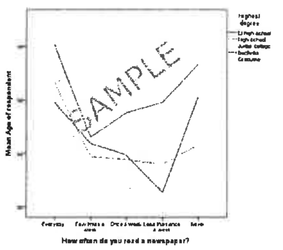
Multiple Line Charts of Means

- Better than clustered bar charts in comparing variables
- To generate multiple line charts of mean *variable A* by *B* and *C*:
  1. **Graphs > legacy dialogs > line**
  2. **>** select **multiple** and **Summaries For Group Of Cases > Define**
  3. Make settings (see red color text below)



Task 3b – Multiple Line Charts of Means

1. Use the previous slide to generate the following clustered line charts

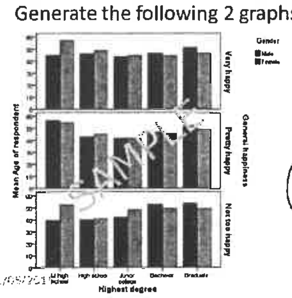


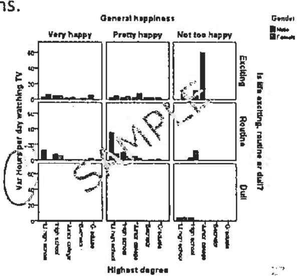
Task 3b answer – Multiple Line Charts of Means

- Clustered line charts of mean *age* by *news* and *degree*:
  1. **Graphs > legacy dialogs > line**
  2. **>** select **multiple** and **Summaries For Group Of Cases > Define**
  3. Set **Category Axis** to news,
  4. Set **Define lines By** to degree,
  5. Set **Line Represent** to other statistics > set **Variable** to age > **Ok**

Assignment 1a

- **Data file**: "gssdata.sav"
- **Statistic tool**: Cluster Bar Chart
- **Requirements** :
  Generate the following 2 graphs.





↳ Use Panel by
Rows: [happy]

Change Statistics
Mar

Assignment 1b

- **Data file**: "gssdata.sav"
- **Statistic tool**: Summary Table, Multiple Line Chart
- **Requirements** : mean mode ||/ median
  1. Generate **central tendency** and **variability** statistics on the variables ***tvhours***, ***degree***, ***happy***, ***life*** and ***richwork***. You may use 1 or 2 tables.
  2. Produce **line charts** using ***tvhours***, ***degree***, ***happy***

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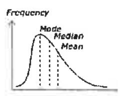
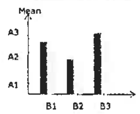
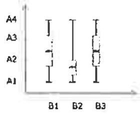
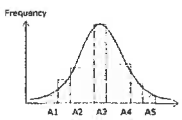
- Normal Distribution



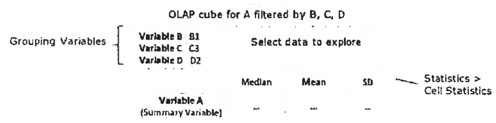






OLAP Cubes

- Use OLAP cube to select subset of data to generate statistics interactively
- To generate OLAP cube for *variable A* filtered by *B*, *C* and *D*:
  1. **Analyze > Reports > OLAP Cubes**
  2. Make settings (see red color text below)
  3. Double Click the OLAP cubes to explore the stats with different variable selections



Task 4 - OLAP Cubes

1. Use the previous slide to generate the following OLAP cube

OLAP Cubes

How often do you read a newspaper? Everyday

Gender Female

Highest degree High school

	Median	Mean	Std. Deviation
Age of respondent	56.50	56.09	18.061

Task 4 answer - OLAP Cubes

- Use OLAP cube to select subset of data to generate statistics interactively
- OLAB cube for *age* filtered by *news*, *sex* and *degree*:
  1. **Analyze > Reports > OLAP Cubes >** set **Summary Variable** to age
  2. Set **Grouping Variables** to news, sex, degree
  3. **> Statistics >** set **Cell Statistics** to Median, Mean, Standard Deviation
  4. **> Continue > OK**
  5. **Double Click** the OLAP cubes to edit >
  6. Set **How Often...?** to every day.
  7. Set **Gender** to female.
  8. Set **Highest Degree** to high school

OLAP Cubes

How often do you read a newspaper? Everyday

Gender Female

Highest degree High school

	Median	Mean	Std. Deviation
Age of respondent	56.00	56.09	18.081

Percentile

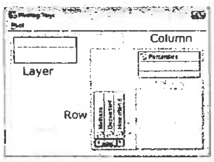
- Generate percentiles for all subsets of data
1. **Analyze > Descriptive Statistics > Explore,**
2. Make setting (see red color text below)
3. Choose **Display > Statistics,**
4. Double Click the percentiles table to use Pivoting Trays to change display of the table

uncheck descriptive

Percentiles of A by B

		Variable A (Dependent Variable)			
		Variable B (Factor List)			
		B1	B2	B3	B4
Percentiles (Statistics)	5	...	...	...	
	10	...	...	...	
	25				
	...	...	...	...	
	95	...	...	...	

Pivoting Trays



Task 5 – Percentile

1. Use the previous slide to generate the following percentiles table
2. From the 75th percentile, how many percent of people who read newspaper every day are older than 68 years?
3. how many percent of people who read newspaper every day are younger than 39 years?
4. How many percent of daily readers are between the age of 39 and 68?
5. What is the interqualtile range of age of people who never read newspaper?

		Age of respondent				
		How often do you read a newspaper?				
		Everyday	Few times a week	Once a week	Less than once a week	Never
Percentiles	5	25.00	20.00	22.00	21.05	22.00
	10	28.00	22.80	23.90	22.10	23.90
	25	39.00	27.00	26.25	28.25	30.50
	50	51.00	39.00	40.50	36.00	45.00
	75	68.00	50.00	52.75	50.00	55.75
	90	76.00	64.00	64.70	65.90	77.00
	95	81.60	72.20	75.00	75.75	81.55

Task 5 answer – Percentile

- Generate percentiles for all subsets of data
- Percentiles of *age* by *news*:
  1. **Analyze > Descriptive Statistics > Explore,**
  2. set **Dependent Variable** to age, set **Factor List** to news
  3. **Set Display** to Statistics
  4. Click **Statistics** > check only Percentiles > **Continue > OK**
  5. **Double Click** the percentiles table to edit > **Pivot > Pivoting Trays**
  6. Set **Layer** to Methods, set **Row Dimension** to Percentiles
  7. Set **Column Dimension** to Dependent variables, how often...?
  8. **Close** the Pivoting Trays and Table Editor

		Age of respondent				
		How often do you read a newspaper?				
		Everyday	Few times a week	Once a week	Less than once a week	Never
Percentiles	5	25.00	20.00	22.00	21.05	22.00
	10	28.00	22.80	23.90	22.10	23.90
	25	39.00	27.00	26.25	28.25	30.50
	50	51.00	39.00	40.50	36.00	45.00
	75	68.00	50.00	52.75	50.00	55.75
	90	76.00	64.00	64.70	65.90	77.00
	95	81.60	72.20	75.00	75.75	81.55

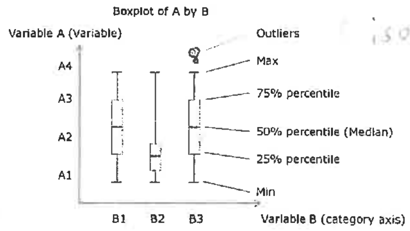
Task 5 answer – Percentile (cont')

1. Use the previous slide to generate the following percentiles table
2. From the 75th percentile, how many percent of people who read newspaper every day are older than 68 years?
   - 25%
3. how many percent of people who read newspaper every day are younger than 39 years?
   - 25%
4. How many percent of daily readers are between the age of 39 and 68?
   - 50%
5. What is the Interqualtile range of age of people who never read newspaper?
   - 30.5 – 55.75

		Age of respondent				
		How often do you read a newspaper?				
		Everyday	Few times a week	Once a week	Less than once a week	Never
Percentiles	5	25.00	20.00	22.00	21.05	22.00
	10	28.00	22.80	23.00	22.10	23.80
	25	39.00	27.00	26.25	28.25	30.50
	50	51.00	39.00	40.50	36.00	45.00
	75	68.00	50.00	52.75	50.00	55.75
	90	76.00	64.00	64.70	65.90	77.00
	95	81.60	72.20	75.00	75.75	81.55

Boxplot

- Simple visualization of the data distribution
- Boxplot of *variable A* by *B:*
  1. **Graphs > Legacy Dialogs > Boxplot**
  2. Select **Simple** icon and **Summaries For Group Of Cases > Define**
  3. Make settings (see red color text below)



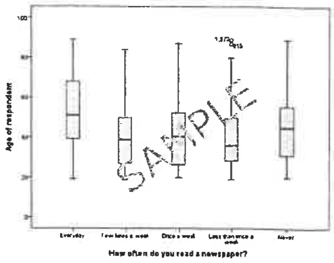
(isolated case)

S.D > 36

S.D < -36

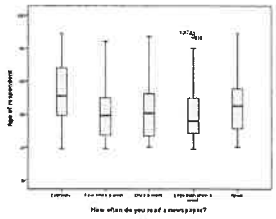
Task 6 - Boxplot

1. Use the previous slide to generate the following Boxplot



Task 6 answer - Boxplot

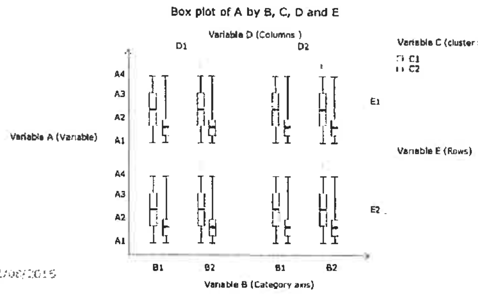
- Simple visualization of the data distribution
- Boxplot of *age* by *news:*
  1. **Graphs > Legacy Dialogs > Boxplot**
  2. Select **Simple** icon and **Summaries For Group Of Cases > Define**
  3. Set **Variable** to <u>age</u>, set **Category Axis** to <u>news</u> **> Ok**



Clustered Box Plot Summaries for Group of Cases

To generate the cluster box plot summaries for group of cases:

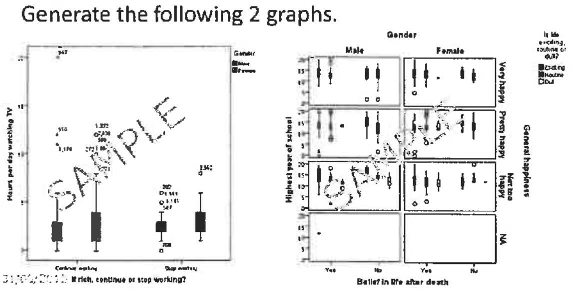
1. **Graphs > Legacy Dialogs > Boxplot >**
2. Select **Cluster** icon and **Summaries For Group Of Cases > Define**
3. Make settings (see red color text below)



Assignment 2a

- **Data file**: "gssdata.sav"
- **Statistic tool**: Box Plot
- **Requirements** :

Generate the following 2 graphs.



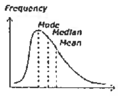
Assignment 2b

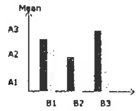
- **Data file**: "gssdata.sav"
- **Statistic tool**: OLAP Cube, Percentiles, Box Plot
- **Requirements** :

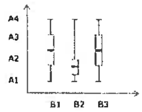
Use the 3 statistic tools to generate **central tendency** and **variability** statistics on the variables ***age***, ***degree*** and ***happy***.

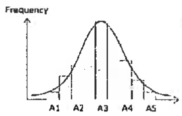
In this session

- What is descriptive statistics?
- Central tendency and Variability
- Generate statistics
  - Summary Table, Pie Chart, Cluster Bar Chart, Cluster Line Chart
  - OLAP Cube, Percentiles, Box Plot
- **Normal Distribution**



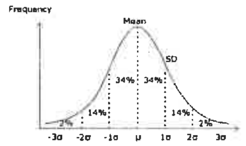






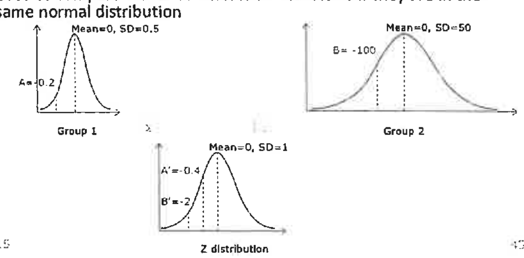
Normal Distribution

- Normal Distribution and Normal Curve
  1. Large number of independent random values tend to cluster around the mean
  2. Symmetrical bell shape
  3. Mean and standard deviation determine the distribution shape
  4. The position of a value in the normal curve determine the frequency or probability of the value



Standard normal distribution and Z score

- Standard normal distribution
  - A normal distribution with **mean=0, SD=1**
- z score (standard score)
  - Transform a normal value to a standardized Z value
  - Used to compare values of different variables as if they are in the same normal distribution



z scores in SPSS

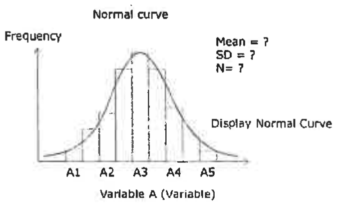
- To calculate z scores for *age* in SPSS:
  1. **Analyze > Descriptive Statistics > Descriptive**
  2. Set **Variable** to age
  3. Check Save standardized values as variables
  4. OK
  5. Check z score in data view
- *Use **Transform > Visual Binning** to find cases with a certain standard score

z score for age in **SPSS data view**

year	Age	...	Zage
...	...	...	.02
...	...	...	-.52
...	...	...	2.17

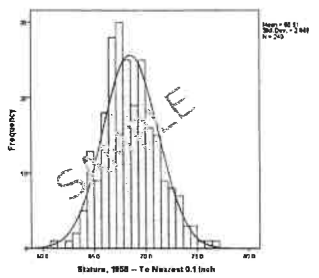
Normal Distribution and Curve

- To generate normal distribution and curve of a *variable A:*
  1. **Graphs > Legacy Dialogs > Histogram**
  2. Make settings (see red color text below)



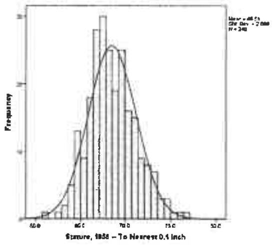
Task 7 - Normal Distribution and Curve

1. Double click file "electric.sav" to open
2. Use the previous slide to generate the following normal distribution and curve



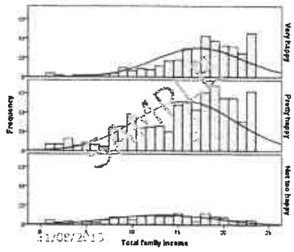
Task 7 answer - Normal Distribution and Curve

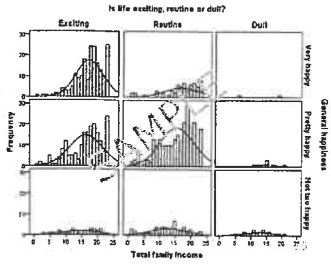
- Generate normal distribution and curve of *ht58:*
  1. **Double click** file "electric.sav" to open
  2. **Graphs > Legacy Dialogs > Histogram**
  3. Set **Variable** to ht58, check Display Normal Curve > **OK**



Assignment 3

- **Data file:** "gssdata.sav"
- **Statistic tool:** Histogram
- **Requirements :**
  1. Try to generate the following 2 graphs.
  2. Make observations on the stats and draw meaningful conclusions.





Conclusion

- Summarize data using descriptive statistics
- Tools to present descriptive statistics
  - Summary table
  - Pie chart, bar chart and line chart
  - OLAP cube, percentiles, box plot
- Normalization of values using z score

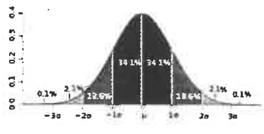
Standard Normal distribution

- z distribution
  - A transformed normal distribution with mean=0, SD=1
  - z score
    - Transform a normal value to a standardized Z value
    - How many standard deviation is the value from the mean
    - Measure individual performance relative to the group performance
    - Compare A's performance in group 1 with B's performance in group2 (the 2 groups may have different ranges and mean)
  - Transform a z score to a percentage between value and mean
  - Raw value ↔ z score ↔ percentage/frequency/probability

$$z = \frac{X - M}{SD}$$

Percentage, Frequency and Probability statements

- Probability
  - Used to describe any uncertainty in science
- Equivalent statements:
  - **34.1%** of all events fall between mean and 1SD
  - **Frequency** of occurrence between mean and 1SD is 34.1%
  - **Probability** of an event between mean and 1SD is 0.341



SPSS frequently used commands

- Analyze > Descriptive Statistics >
  - Frequencies (Frequency table)
  - Crosstabs (Crosstabulation)
  - Explore (Stem-and-leaf plots, percentile)
- Analyze > Compare Means >
  - Means (single/multi-variable Summary Statistic Table)
- Analyze > Reports > OLAP Cubes
- Graphs > legacy dialogs >
  - Pie Chart
  - Bar > Simple (Bar chart)
  - Bar > Clustered > Summaries For Group Of Cases (Clustered bar chart of frequency/mean)
  - Histogram (Normal Curve)
  - Line > Multiple > Summaries For Group Of Cases (Clustered line charts of frequency/mean)
  - Boxplot
- Data > Select Cases >
  - If Condition Is Satisfied
  - All Cases > Ok

SPSS Frequently used command settings

- **main dimension:**
  - variable, category list, category axis, slice by, rows/columns, independent list, grouping variable, factor list
- **other dimension:**
  - penal by, cluster by, layer1, Change Statistic > check Percentage Inside
- **variable:**
  - displayed list, Bars Represent, dependent list, summary variable, dependable variable, other statistics
- **statistic:**
  - count, percentage, sum, mean, median, stand. deviation, stand. error, variance